

Pengxiang (Eric) Wang

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🌐 ek-flowity.site

SUMMARY

- Third-year Computer Science student at the University of Toronto specializing in training and optimizing deep learning models for efficient inference and generalization. Experienced in model distillation, speech synthesis, and LLM-based retrieval systems with measurable improvements in accuracy and latency. Strong foundation in PyTorch, Transformer architectures, and experimental analysis. Seeking ML Engineer Internship.

EDUCATION

- **University of Toronto** Toronto, ON
• *Bachelor of Science in Computer Science (Specialist); GPA: 3.89/4.00* *Sep. 2023 – May. 2028 (Expected)*

PROJECTS

- **Expressive and Efficient TTS via Model Distillation (PyTorch, Transformer)** *Dec. 2025*
 - Designed and trained a 0.5B phoneme-based Transformer TTS model distilled from a 3B teacher to optimize latency-accuracy tradeoff.
 - Implemented knowledge distillation with temperature scaling and masking-based auxiliary loss to disentangle prosody from speaker identity.
 - Conducted ablation studies on masking strategies and distillation hyperparameters, improving training stability and reducing WER variance across unseen speakers.
 - Achieved 2.89% WER (clean) and 2.16× faster inference compared to 3B baseline, maintaining real-time deployment feasibility.
 - Generated 33+ hours of high-fidelity synthetic speech across 7 emotional categories; performed data filtering and phoneme balance validation.
 - **Open Source:** github.com/BarryFutureman/NeuTTS-Express
- **LLM Agent with RAG Optimization (Spring AI, PGVector)** *Sep. 2024*
 - Built a production LLM system integrating retrieval-augmented generation and multi-model routing for structured reasoning tasks.
 - Designed a PGVector-based semantic retrieval pipeline with custom ranking strategies; improved retrieval precision by 45%.
 - Implemented streaming inference (SSE) and asynchronous execution to reduce time-to-first-token by 60% and increase concurrency by 2.5×.
 - Integrated local and cloud LLM backends for cost-efficient inference and privacy control.
 - **Live Demo:** ek-flowity.site/projects/ai-agent/

SKILLS

- **AI/ML:** PyTorch, Transformers, Model Distillation, LLM Fine-Tuning, RAG, TTS
- **Systems:** REST APIs, Redis, SSE, Distributed Systems
- **ML Concepts:** Optimization (SGD/Adam), Regularization, Generalization, Evaluation Metrics
- **Cloud/Tools:** Docker, Nginx, Linux, Git, PGVector
- **Language:** Python, Java, TypeScript, SQL